

AHMED BARGOUGUI

AI & Data Engineering | Backend Developer | Computer Science — Final Year

+216 95 430 441 • hamoudabargougui@gmail.com • Sidi Bouzid, Tunisia • [linkedin.com/in/ahmedbargougui](https://www.linkedin.com/in/ahmedbargougui) • github.com/Ahmedbrg

PROFILE

Final-year Computer Science student specializing in AI engineering and backend systems, with two internships at Sagemcom — building a RAG- and LLM-based Generative AI system for telecom network validation, and an ETL/BI pipeline for KPI reporting. Builder of production-grade AI applications (multi-agent risk analysis, multimodal video search) and backend systems serving 32,000+ users. Seeking a PFE internship in AI Engineering or Backend Engineering.

PROFESSIONAL EXPERIENCE

- Telecom Data & AI Engineering Intern** — Sagemcom · Ben Arous, Tunisia07/2026 – 08/2026
 - Built an ICP/MTP validation system that extracts data via OCR from telecom compliance documents and generates automated PASS/FAIL reports.
 - Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline with LLM-based analysis to automatically interpret ICP/MTP validation results, including prompt design and token/context management for reliable, cost-controlled LLM calls.
 - Layered a CrewAI multi-agent system on top of the validation output to generate automated, human-readable interpretation reports.
 - Engineered an end-to-end n8n automation flow — validation → Generative AI analysis → production webhook → automated Gmail reporting — removing manual review from the compliance workflow.
- Business Intelligence Engineer Intern** — Sagemcom · Ben Arous, Tunisia07/2025 – 08/2025
 - Cleaned and prepared raw CSV data and validated data requirements ahead of ingestion into the reporting pipeline.
 - Developed Python scripts for backend-oriented data processing and automation, focusing on understanding the full system flow from raw data to backend processing.
 - Designed relational database schemas and a Star Schema data warehouse; built SQL Server-backed ETL pipelines with Pandas and SQLAlchemy.
 - Standardised 10,000+ records via Power Query and built Power BI dashboards for real-time KPI tracking.

PROJECTS

- AI Risk Analyst — Enterprise Risk Platform** — github.com/Ahmedbrg/Risk_Analyst2026
 - Built a full-stack enterprise risk platform (FastAPI + React) powered by a 5-agent CrewAI system that analyzes risk narratives end to end.
 - Implemented Retrieval-Augmented Generation with Qdrant as the vector store to ground agent outputs in relevant reference material.
 - Designed a deterministic, ISO 31000-aligned scoring engine so risk scores are reproducible rather than purely LLM-generated.
 - Automated the reporting workflow with n8n and added a QC-gated PDF export step to ensure only validated reports are released.

Stack: Python · FastAPI · CrewAI · Qdrant · RAG · React · n8n · Docker
- ScanVD — AI Video Search Engine** — github.com/Ahmedbrg/ScanVD2026
 - Built a multi-model AI pipeline (Whisper + YOLOv8 + CLIP) that extracts precise timestamps from natural-language queries over video content.
 - Deployed a live demo on HuggingFace Spaces for public testing.

Stack: Python · Whisper · YOLOv8 · CLIP · HuggingFace · Docker
- Other Projects**
 - Forsa — Tunisian Mobile Super-App:** dark-themed React Native super-app (community feed, marketplace, freelance jobs, messaging) with a JWT-secured Express/MongoDB API. github.com/Ahmedbrg/Forsa
 - Secure Auth System — Gaming Community:** secure REST API (Node.js, Express, MySQL) and C# desktop launcher for a 32,000-member community — HWID binding, OAuth2, JWT, rate limiting.

TECHNICAL SKILLS

AI / GenAI: RAG · LLMs · CrewAI · Qdrant · Whisper · YOLOv8 · CLIP · OCR · Embeddings
Automation: n8n · Webhooks · Gmail API · Workflow Automation · JSON Pipelines
Backend: Node.js · Express · FastAPI · REST APIs · JWT · OAuth2 · MySQL · MongoDB · C#
Data / BI: SQL Server · ETL · Pandas · SQLAlchemy · Power BI · Star Schema
Languages: Python · JavaScript · TypeScript · Java · C++ · C# · SQL
Tools: Git · Docker · Linux · Postman · VS Code · Jupyter

EDUCATION • LEADERSHIP • LANGUAGES

- Bachelor's Degree in Computer Science** — FSTSBZ, University of Kairouan09/2024 – 06/2027 · Final Year (PFE)
 - Coursework: Algorithms · Data Structures · Databases · OOP · OS · Networks · AI.
 - AgriLink — AI farming app built at the TGN Hackathon 2025 (React Native).
 - Founder & President, Club DevUp — organised AI/ML and career workshops for 50+ students (11/2025 – 05/2026).
 - Trainer & Jr. Secretary General, JCI Sidi Bouzid (2023 – Present).
 - Ranked 3rd, ACM-style Competitive Programming Contest. Languages: Arabic (Native) · French (Professional) · English (Professional).